Practical-1

Problem Definition:

Design a 4-input arithmetic-control logic function H(A,B,C,D) = 1 if the 4-bit input (A B C D) matches any of six specific patterns(select by yourself; otherwise H = 0).

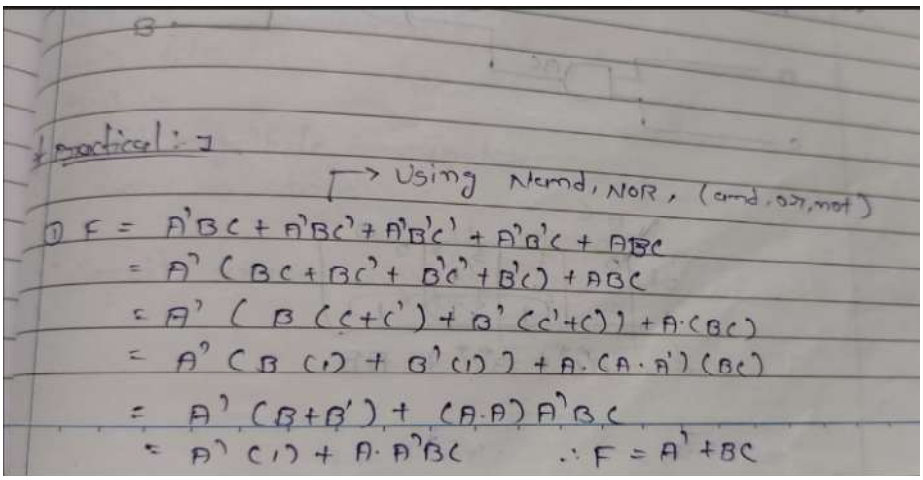
You must:

1. Choose patterns.
2. Derive the Sum-of-Products form (explicit minterms).
3. Attempt simplification with a Karnaugh map and with Boolean algebra; explain differences.
4. Decide which implementation style (NAND-only, NOR- only, or mixed) is most suitable and explain why.
5. Implement the final design in TinkerCad and demonstrate operation with test vectors.
6. Convert the current SOP to a minimal Product-of-Sums (POS) and compare NOR-only implementation cost with NAND-only.

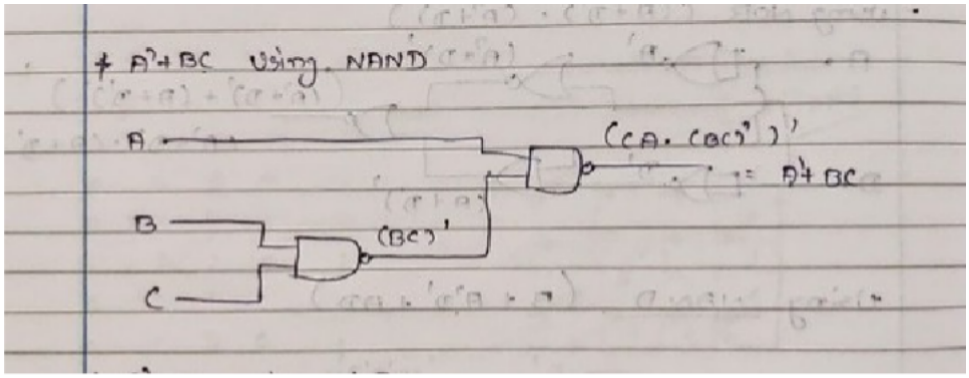
Tools/Technology To Be Used:

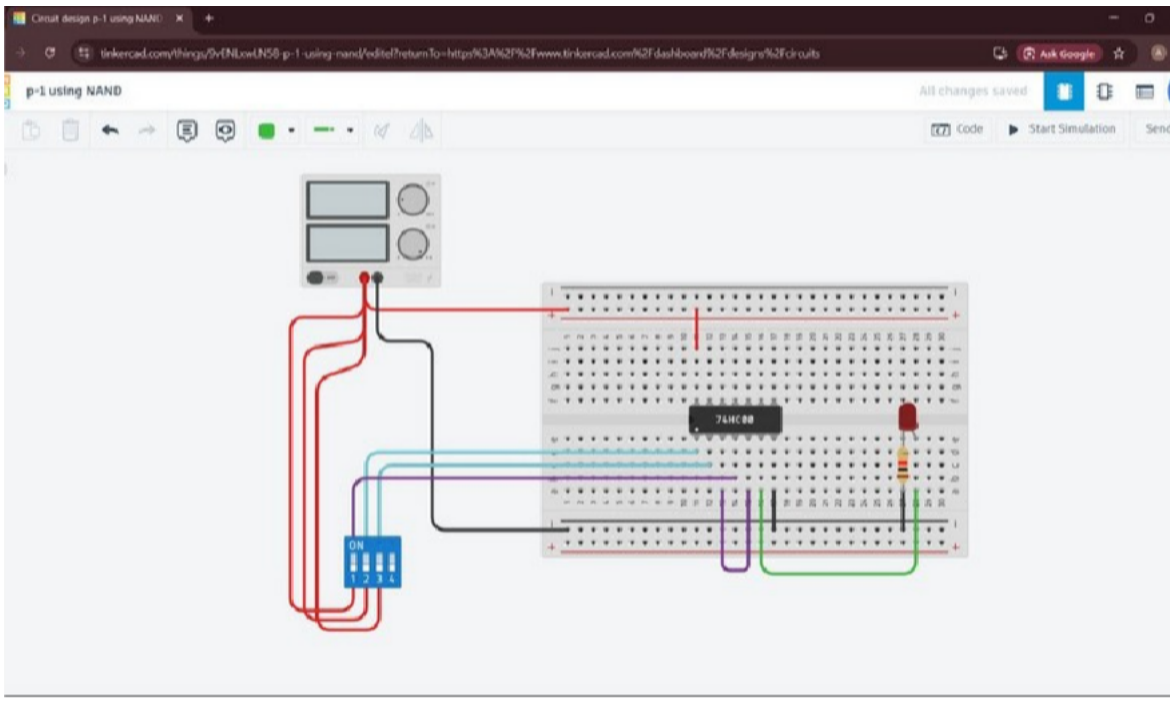
- Logic design: pen and paper
- Gate simulation & wiring: TinkerCad Circuits
- Optional: Multi-input gate ICs or build using cascaded 2-input gates.

1) simplification with Boolean algebra:

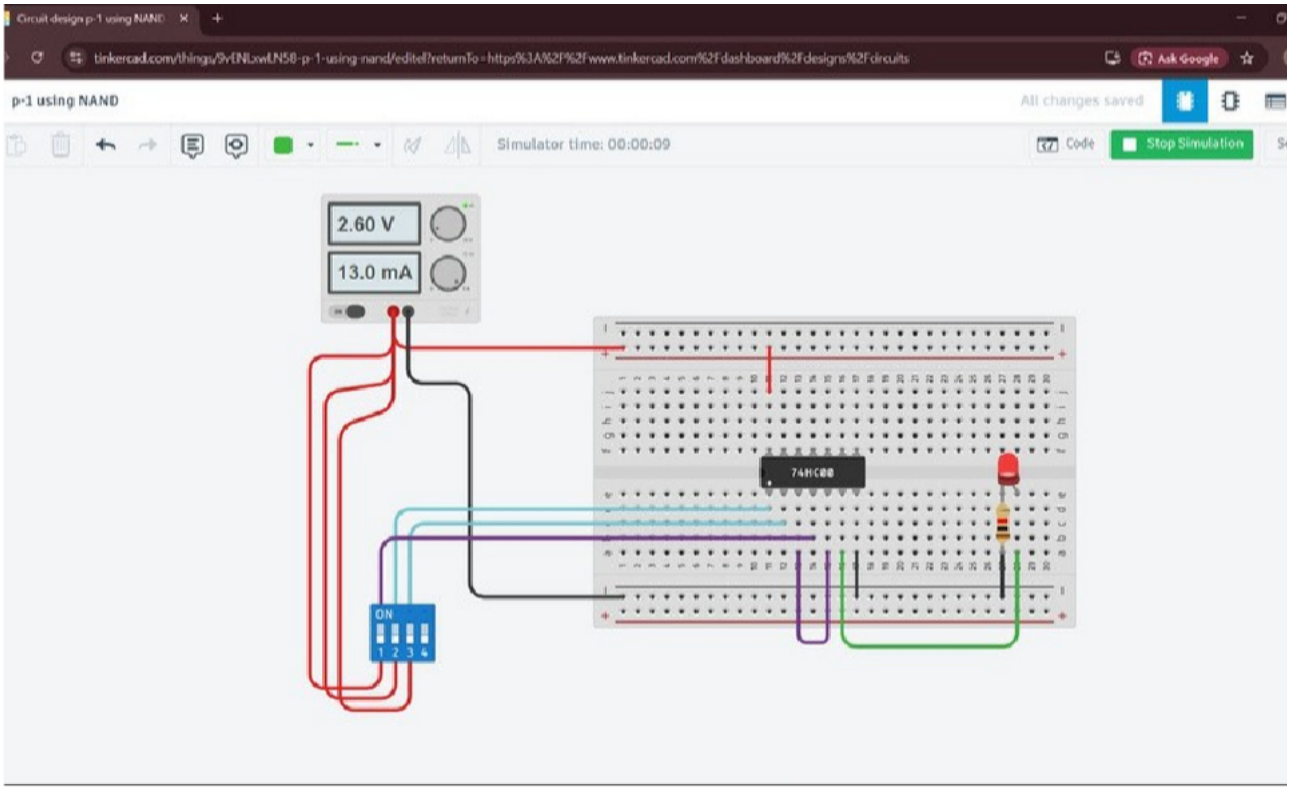


1) Using Nand gate:

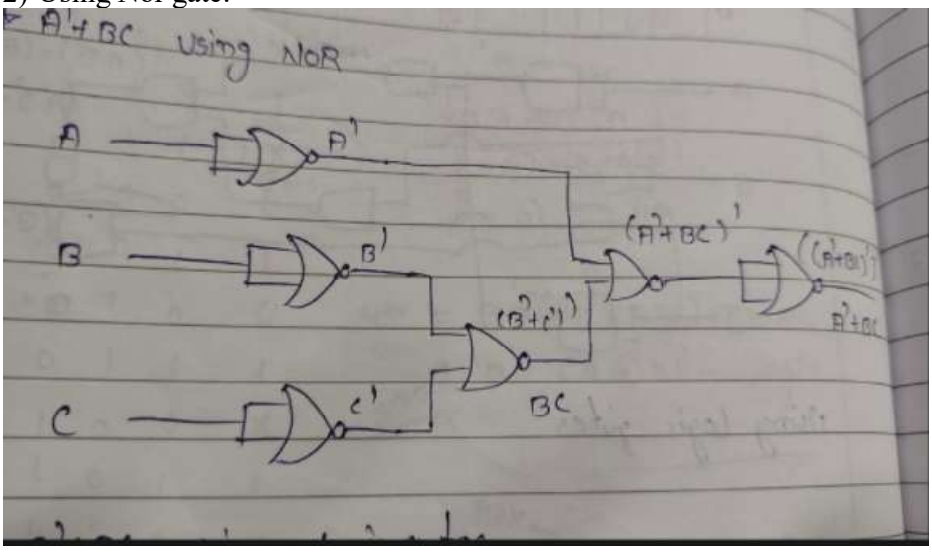


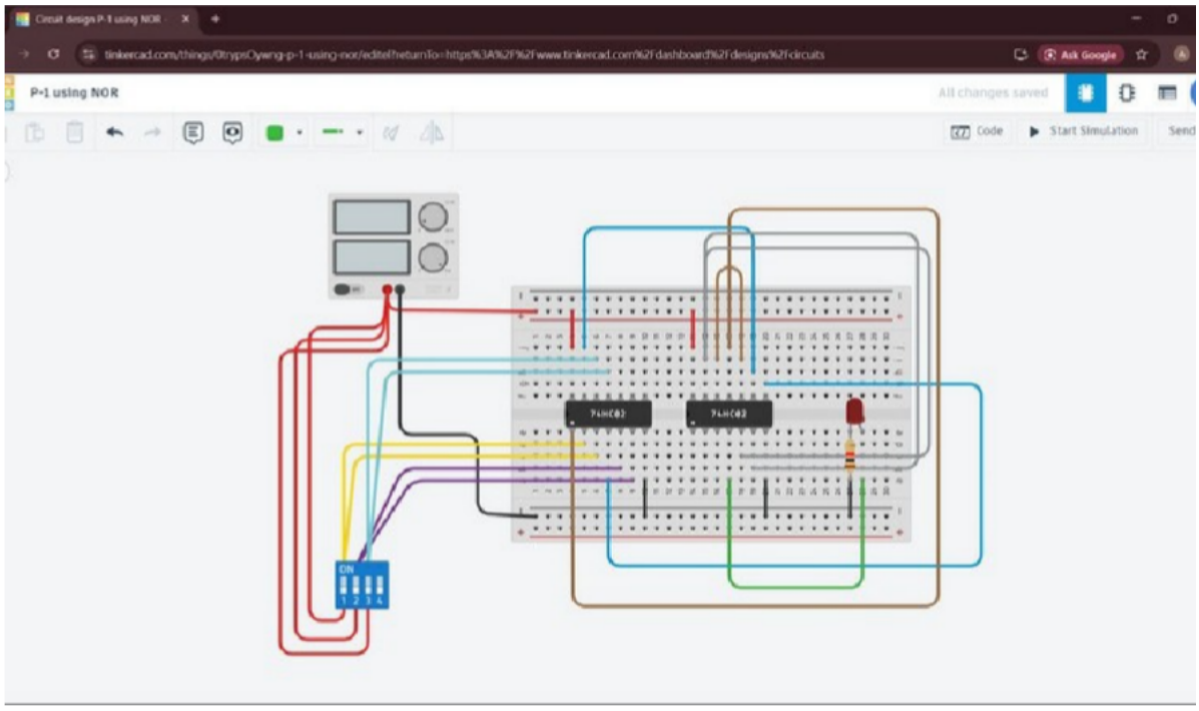


Output:

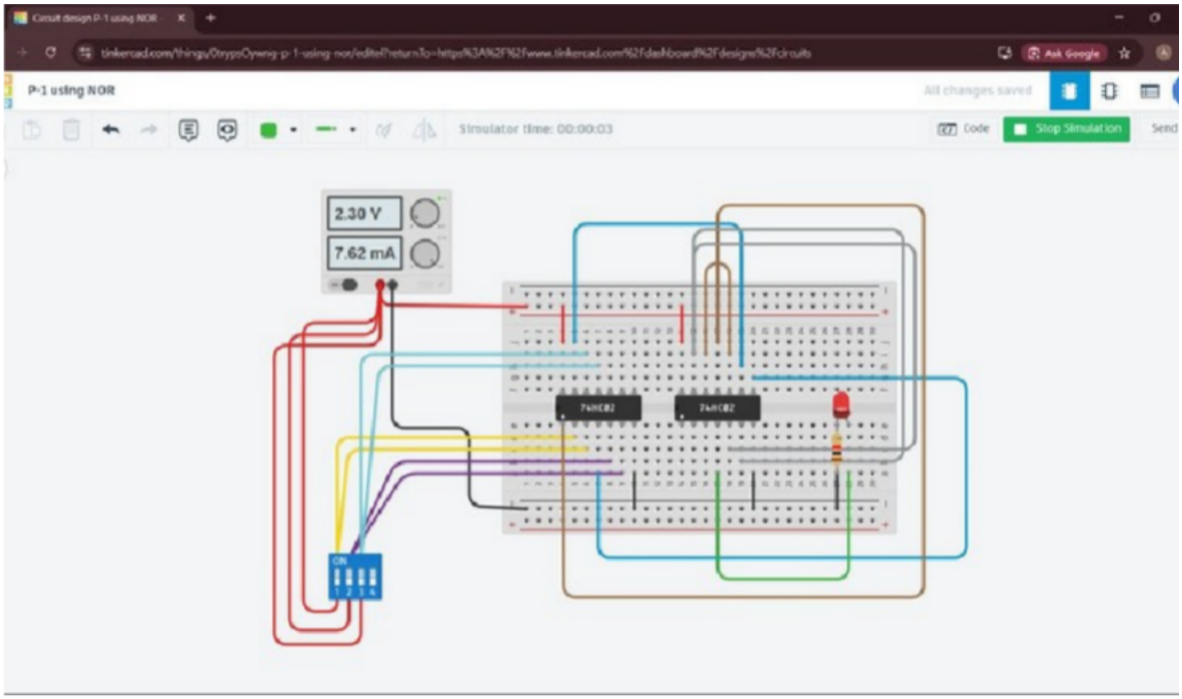


2) Using Nor gate:

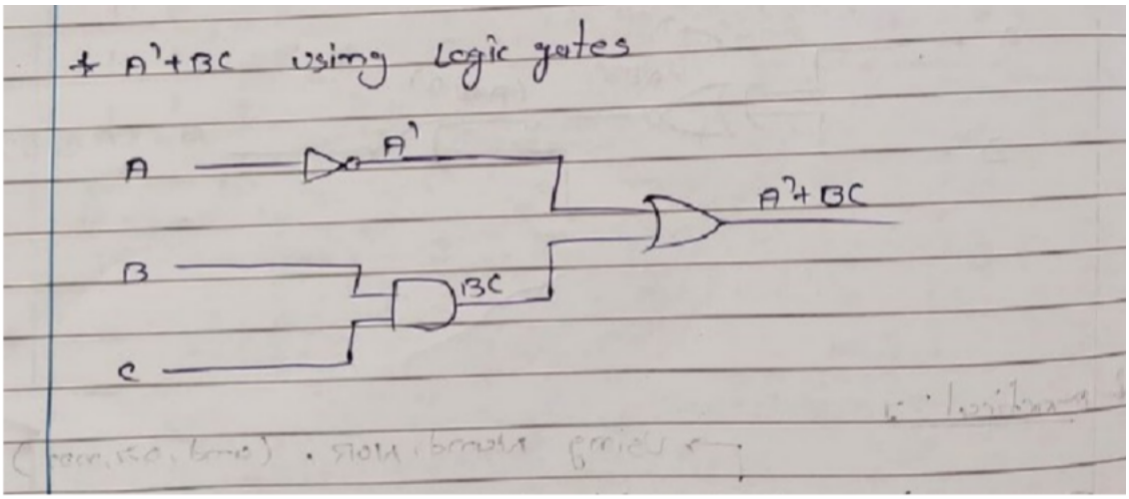


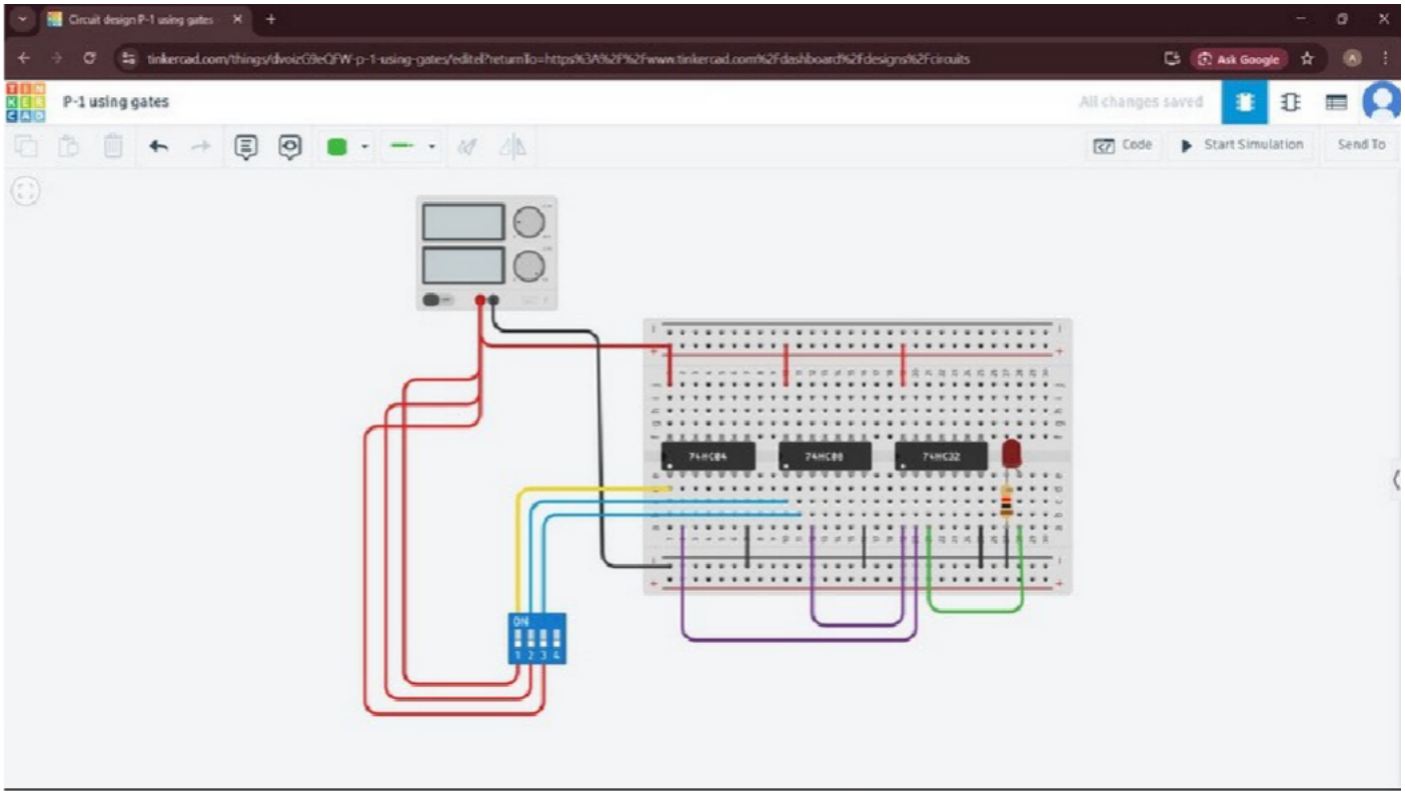

P=1 using NOR
All changes saved
Code
Start Simulation
Send

Output:

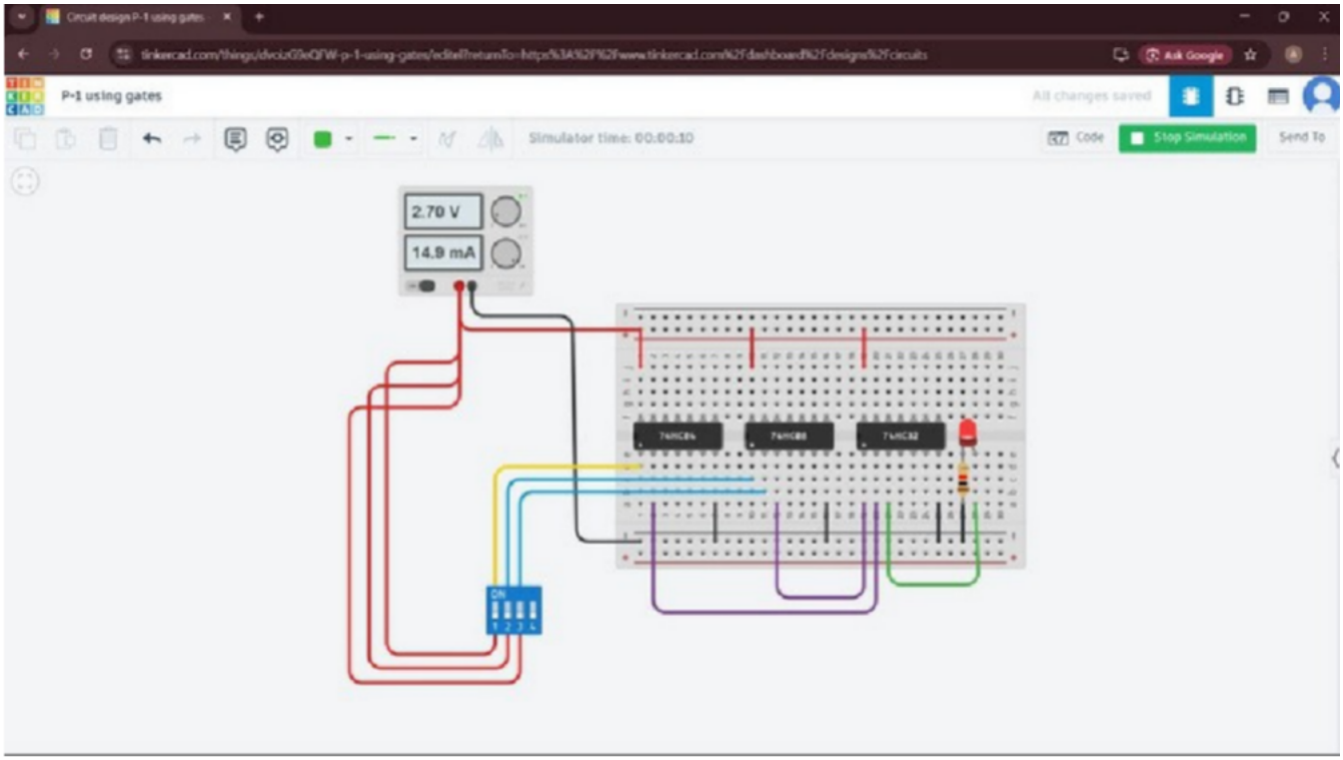

P=1 using NOR
All changes saved
Simulator time: 00:00:03
Code
Stop Simulation
Send
2.30 V
7.62 mA

3) Using And ,Or, Not gate:

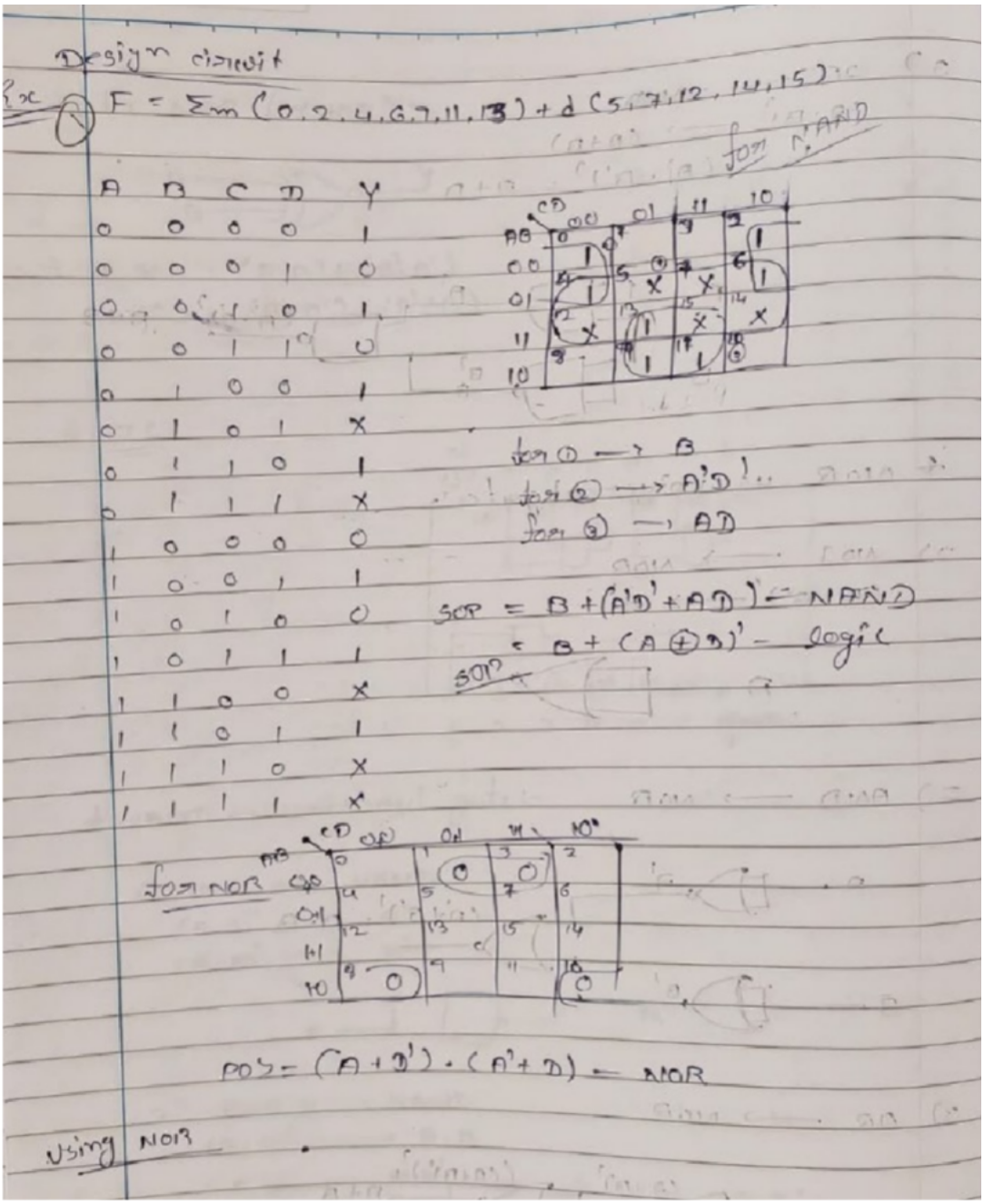




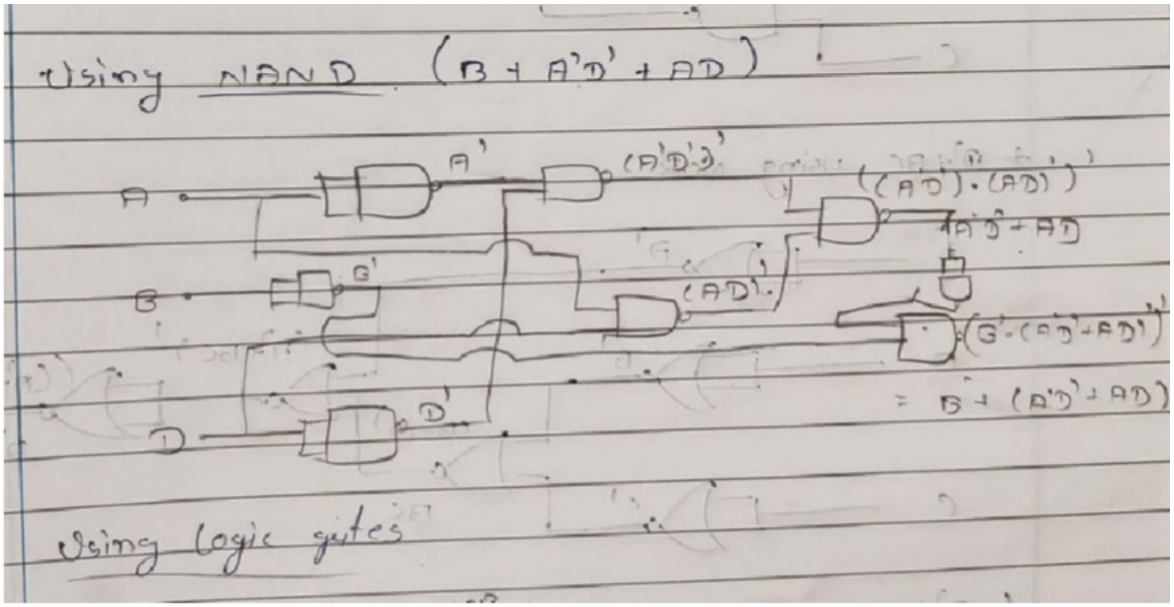
Output:


P-1 using gates
Simulator time: 00:00:10
Code
Stop Simulation
Send To
2.70 V
14.9 mA

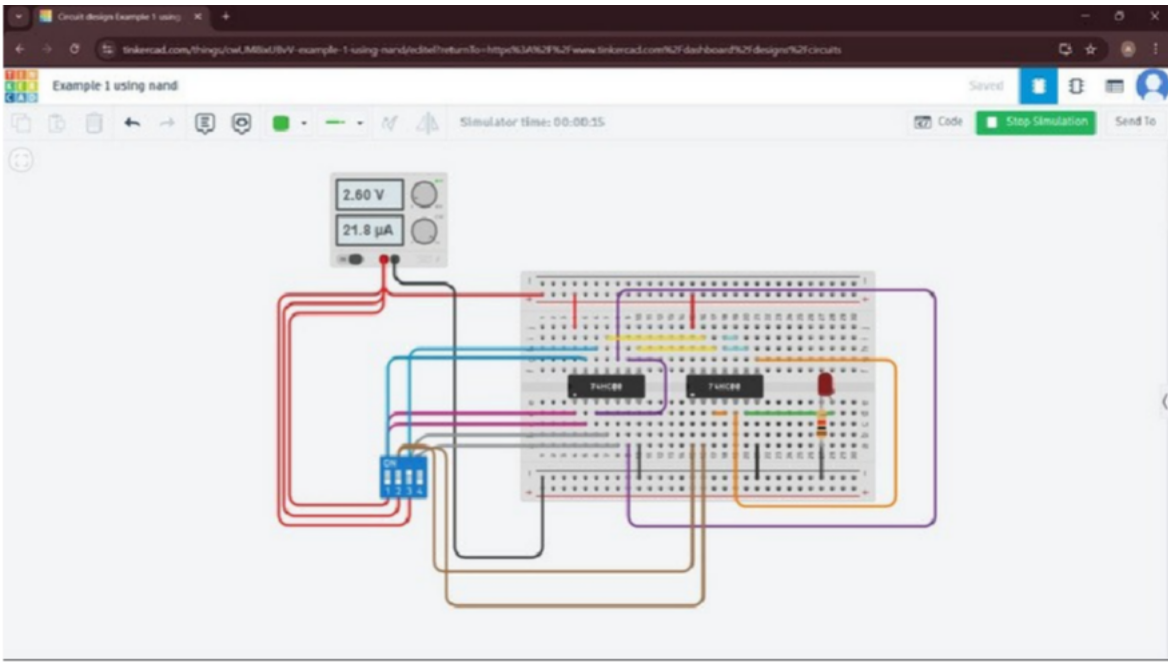
2) simplification with a Karnaugh map:

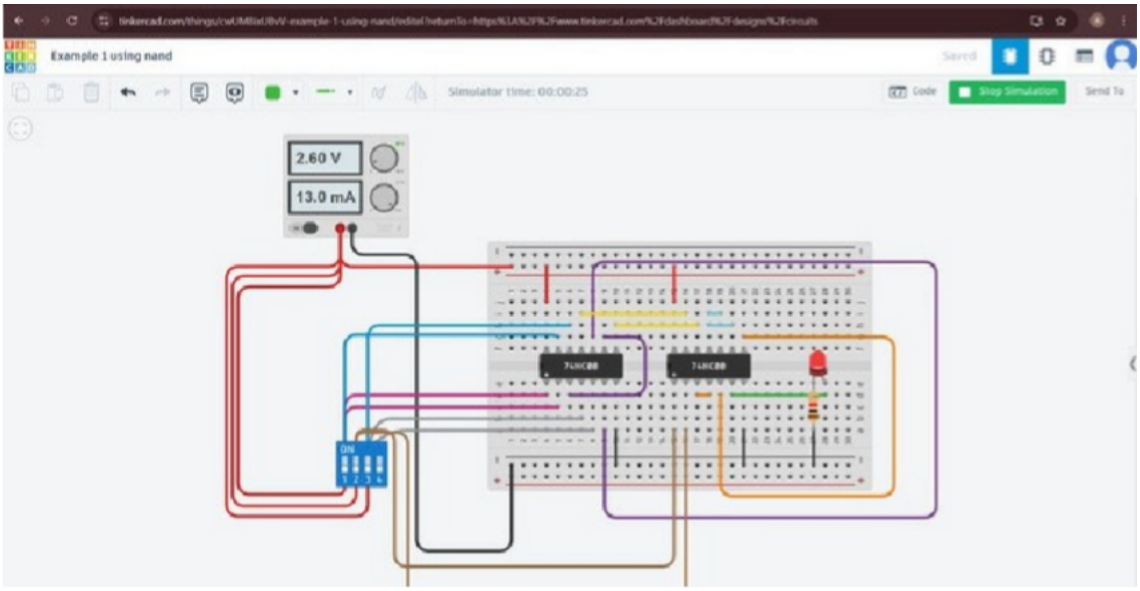


1) Using Nand gate:

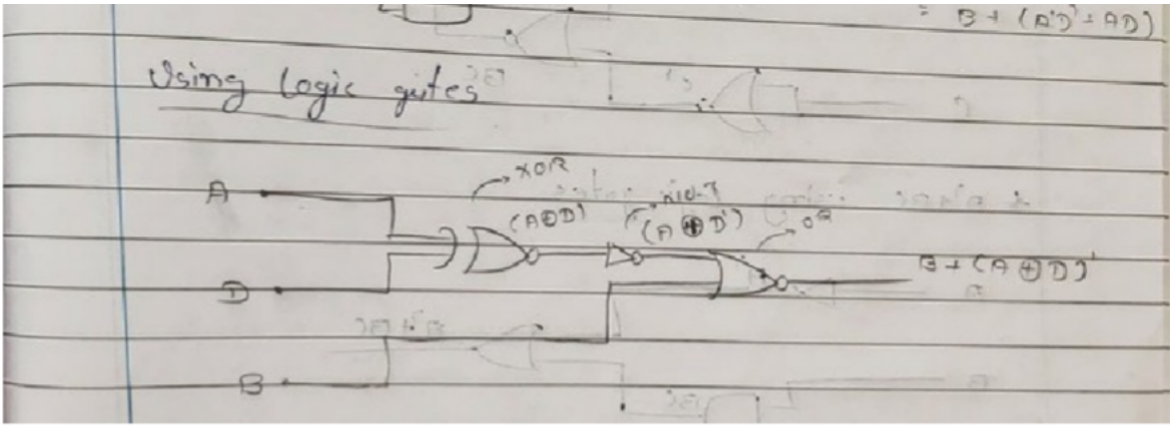


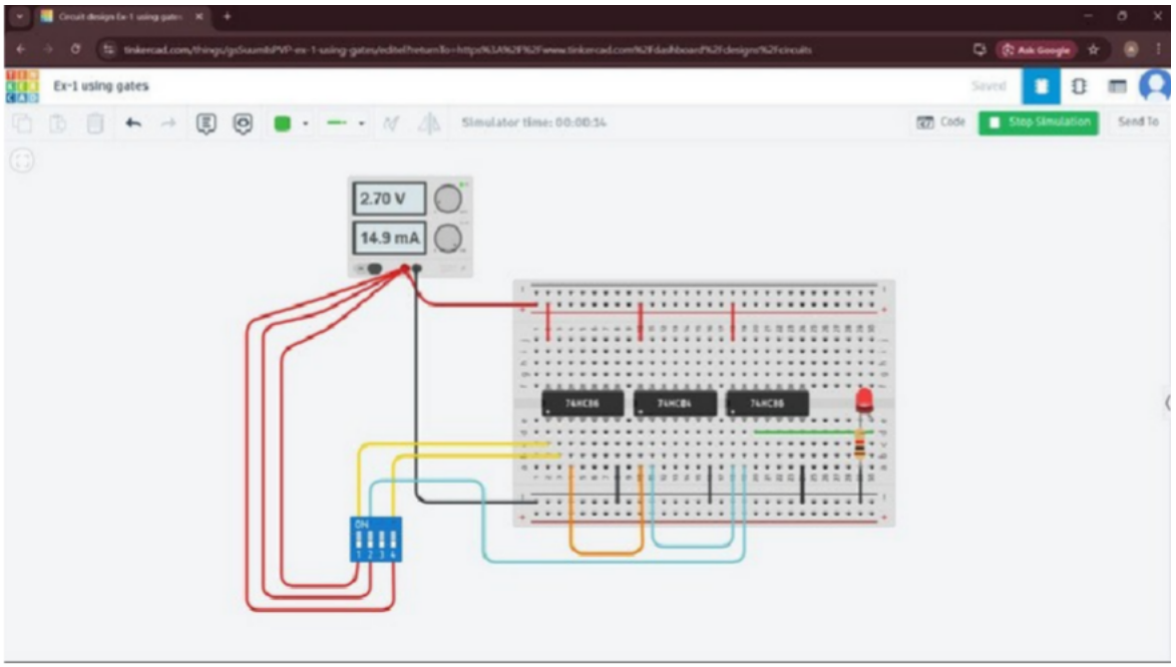
Output:





2) Using Logic gates:




Ex-1 using gates
2.70 V
14.9 mA

Conclusion: This practical successfully demonstrates the process of transforming a specific set of input patterns into an optimized hardware logic function using Karnaugh Map (K-map) and Boolean algebraic techniques. By comparing SOP (Sum-of-Products) and POS (Product-of-Sums) forms, the design reveals how the choice of implementation (NAND-only vs. NOR-only) directly impacts the total gate count and circuit complexity. The transition from theoretical minterms to a physical TinkerCad simulation confirms that Boolean minimization preserves the functional integrity of the original truth table. Ultimately, the project validates that strategic logic reduction not only simplifies the schematic but also reduces the physical hardware cost required to detect specific 4-bit arithmetic control sequences.